

System of linear Equations & their nature of solution

Consider the system of linear equations as below:-

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \dots + a_{2n}x_n = b_2$$

$$\vdots \quad \vdots \quad \vdots \quad \ddots \quad \vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \dots + a_{mn}x_n = b_m$$

This system of linear equations can be visualized in the form of a matrix as $AX = b$

Then the system is said to be consistent if:

$$\rho(A:B) = \rho(A)$$

Else, the system is termed as inconsistent.

A system possess a solution only when it is consistent.

Nature of solution: If $[\rho(A:B) = \rho(A)] = n$ then the system has a unique solution.

$$\therefore \text{If } [\rho(A:B) = \rho(A)] < n$$

then the system has infinitely many solution.

Note: ρ : denotes rank of the matrix which is the number of non-zero rows in the matrix.

MODULE - 2

Solution of linear System - Direct and Iterative methods

Matrix Partition Method:

Partition method is used to find the inverse of a matrix. Let the coefficient matrix A be partitioned

$$\text{as } A = \begin{bmatrix} B & C \\ E & D \end{bmatrix}$$

where, B is an $r \times r$ matrix

C is an $r \times s$ matrix

D is an $s \times s$ matrix

E is an $s \times r$ matrix

Here, r and s are positive integers with $r + s = n$

Further, we also assume that A^{-1} is partitioned

$$\text{similarly as } A^{-1} = \begin{bmatrix} X & Y \\ Z & V \end{bmatrix}$$

where, X , Y , V and Z are matrices of the same orders as of B , C , D and E respectively.

Then, we have

$$AA^{-1} = \left[\begin{array}{c|c} B & C \\ \hline E & D \end{array} \right] \left[\begin{array}{c|c} X & Y \\ \hline Z & V \end{array} \right] = \left[\begin{array}{c|c} I_1 & 0 \\ \hline 0 & I_2 \end{array} \right] \rightarrow (*)$$

where I_1 and I_2 are identity matrices of orders r and s respectively.

Thus, from (*) we have :

$$BX + CZ = I_1 \rightarrow (1)$$

$$BY + CV = 0 \rightarrow (2)$$

$$EX + DZ = 0 \rightarrow (3)$$

$$EY + DV = I_2 \rightarrow (4)$$

From (2), Consider $BY + CV = 0$

$$\Rightarrow BY = -CV$$

$$\Rightarrow \boxed{Y = -B^{-1}CV} \rightarrow (5)$$

Substituting (5) in (4), we get :

$$EY + DV = I_2$$

$$E(-B^{-1}CV) + DV = I_2$$

$$(D - EB^{-1}C)V = I_2$$

$$V = (D - EB^{-1}C)^{-1}I_2$$

$$\boxed{V = (D - EB^{-1}C)^{-1}} \rightarrow (6)$$

From (1), Consider $BX + CZ = I_1$

$$\Rightarrow BX = I_1 - CZ$$

$$\Rightarrow X = B^{-1}(I_1 - CZ)$$

$$\Rightarrow \boxed{X = B^{-1} - B^{-1}CZ} \rightarrow (7)$$

Substituting (7) in (3), we get:

$$EX + DZ = 0$$

$$E(B^{-1} - B^{-1}CZ) + DZ = 0$$

$$EB^{-1} - EB^{-1}CZ = -DZ$$

$$EB^{-1} = EB^{-1}CZ - DZ$$

$$EB^{-1} = (EB^{-1}C - D)Z$$

$$- EB^{-1} = (D - EB^{-1}C)Z$$

$$- (D - EB^{-1}C)^{-1} EB^{-1} = Z$$

$$\boxed{- VEB^{-1} = Z} \rightarrow (8)$$

It is obvious that we need to find two inverse matrices B^{-1} and $(D - EB^{-1}C)^{-1}$ of orders σ and δ respectively.

① Use Matrix Partition Method to find the solution of the system of equations:

$$x_1 + x_2 + x_3 = 1$$

$$4x_1 + 3x_2 - x_3 = 6$$

$$3x_1 + 5x_2 + 3x_3 = 4$$

Soln: Let the matrix A be partitioned as:

$$A = \left[\begin{array}{cc|c} 1 & 1 & 1 \\ 4 & 3 & -1 \\ \hline 3 & 5 & 3 \end{array} \right] = \left[\begin{array}{c|c} B & C \\ \hline E & D \end{array} \right] \quad \& \quad A^{-1} = \left[\begin{array}{c|c} x & y \\ \hline z & v \end{array} \right]$$

$$\text{Now, } B^{-1} = \frac{1}{|B|} \text{adjoint}(B)$$

$$\text{adjoint}(B) = \text{Transpose of cofactor matrix}(B)$$

$$|B| = -1$$

$$\text{cofactor matrix}(B) = \begin{pmatrix} 3 & -4 \\ -1 & 1 \end{pmatrix}$$

$$\therefore \text{adjoint}(B) = \begin{pmatrix} 3 & -1 \\ -4 & 1 \end{pmatrix}$$

$$\Rightarrow B^{-1} = - \begin{pmatrix} 3 & -1 \\ -4 & 1 \end{pmatrix}$$

Now, $D - EB^{-1}C$

$$= 3 + [3 \ 5] \begin{bmatrix} 3 & -1 \\ -4 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$= -10$$

W.K.T $V = (D - EB^{-1}C)^{-1}$

$$\therefore V = -\frac{1}{10}$$

Next, $Y = -B^{-1}CV$

$$= \begin{bmatrix} 3 & -1 \\ -4 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \left(-\frac{1}{10}\right)$$

$$= -\frac{1}{10} \begin{bmatrix} 4 \\ -5 \end{bmatrix}$$

Next, $Z = -VEB^{-1}$

$$= -\frac{1}{10} [3 \ 5] \begin{bmatrix} 3 & -1 \\ -4 & 1 \end{bmatrix}$$

$$= -\frac{1}{10} [-11 \ 2]$$

Finally, $X = B^{-1} - B^{-1}CZ$

$$= \begin{bmatrix} -3 & 1 \\ 4 & -1 \end{bmatrix} - \frac{1}{10} \begin{bmatrix} 3 & -1 \\ -4 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \begin{bmatrix} -11 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} -3 & 1 \\ 4 & -1 \end{bmatrix} - \frac{1}{10} \begin{bmatrix} -44 & 8 \\ 55 & -10 \end{bmatrix}$$

$$= \begin{bmatrix} 1.4 & 0.2 \\ -1.5 & 0 \end{bmatrix}$$

Thus, $A^{-1} = \begin{bmatrix} x & | & y \\ \hline z & | & v \end{bmatrix} = \begin{bmatrix} 1.4 & 0.2 & -0.4 \\ -1.5 & 0 & 0.5 \\ 1.1 & -0.2 & -0.1 \end{bmatrix}$

The solution of the given system of equations is: -

$$X = A^{-1}b$$

$$= \begin{bmatrix} 1.4 & 0.2 & -0.4 \\ -1.5 & 0 & 0.5 \\ 1.1 & -0.2 & -0.1 \end{bmatrix} \begin{bmatrix} 1 \\ 6 \\ 4 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 0.5 \\ -0.5 \end{bmatrix}$$

$\therefore x_1 = 1 ; x_2 = 0.5 \text{ \& } x_3 = -0.5$

② Find the inverse of the matrix

$$A = \begin{bmatrix} 2 & 1 & 1 & -2 \\ 4 & 0 & 2 & 1 \\ 3 & 2 & 2 & 0 \\ 1 & 3 & 2 & -1 \end{bmatrix}$$

using partition method and hence solve the system of equations $Ax = b$, where $b = [-10, 8, 7, -5]^T$.

Solⁿ: We write

$$A = \left[\begin{array}{cc|cc} 2 & 1 & 1 & -2 \\ 4 & 0 & 2 & 1 \\ \hline 3 & 2 & 2 & 0 \\ 1 & 3 & 2 & -1 \end{array} \right] = \left[\begin{array}{c|c} B & C \\ \hline E & D \end{array} \right] ; A^{-1} = \left[\begin{array}{c|c} X & Y \\ \hline Z & V \end{array} \right]$$

$$\text{We find that } B = \begin{bmatrix} 2 & 1 \\ 4 & 0 \end{bmatrix} ; B^{-1} = \frac{1}{4} \begin{bmatrix} 0 & 1 \\ 4 & -2 \end{bmatrix}$$

$$\begin{aligned} D - EB^{-1}C &= \begin{bmatrix} 2 & 0 \\ 2 & -1 \end{bmatrix} - \frac{1}{4} \begin{bmatrix} 3 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 4 & -2 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix} \\ &= \frac{1}{4} \begin{bmatrix} 2 & 17 \\ 6 & 25 \end{bmatrix} \end{aligned}$$

$$V = (D - EB^{-1}C)^{-1} = \frac{1}{13} \begin{bmatrix} -25 & 17 \\ 6 & -2 \end{bmatrix}$$

$$Y = -B^{-1}CV = -\frac{1}{52} \begin{bmatrix} 0 & 1 \\ 4 & -2 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} -25 & 17 \\ 6 & -2 \end{bmatrix}$$

$$= \frac{1}{13} \begin{bmatrix} 11 & -8 \\ 15 & -5 \end{bmatrix}$$

$$Z = -VEB^{-1} = -\frac{1}{52} \begin{bmatrix} -25 & 17 \\ 6 & -2 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 4 & -2 \end{bmatrix}$$

$$= \frac{1}{13} \begin{bmatrix} -1 & 15 \\ -6 & -1 \end{bmatrix}$$

$$X = B^{-1} - B^{-1}CZ = \frac{1}{4} \begin{bmatrix} 0 & 1 \\ 4 & -2 \end{bmatrix} - \frac{1}{52} \begin{bmatrix} 0 & 1 \\ 4 & -2 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} -1 & 15 \\ -6 & -1 \end{bmatrix}$$

$$= \frac{1}{13} \begin{bmatrix} 2 & -4 \\ -2 & -9 \end{bmatrix}$$

Hence, we obtain: $A^{-1} = \frac{1}{13} \begin{bmatrix} 2 & -4 & 11 & -8 \\ -2 & -9 & 15 & -5 \\ -1 & 15 & -25 & 17 \\ -6 & -1 & 6 & -2 \end{bmatrix}$

And $X = A^{-1}b = \frac{1}{13} \begin{bmatrix} 2 & -4 & 11 & -8 \\ -2 & -9 & 15 & -5 \\ -1 & 15 & -25 & 17 \\ -6 & -1 & 6 & -2 \end{bmatrix} \begin{bmatrix} -10 \\ 8 \\ 7 \\ -5 \end{bmatrix} = \begin{bmatrix} 5 \\ 6 \\ -10 \\ 8 \end{bmatrix}$

Cholesky Method:

This method is also known as the square root method. If the coefficient matrix A is symmetric and positive definite, then the matrix A can be decomposed as $A = LL^T \rightarrow (1)$

where $L = (l_{ij})$, $l_{ij} = 0$, $i < j$.

Alternatively, A may be decomposed as $A = UU^T$

From (1), the system $AX = b$ becomes

$$LL^T X = b \rightarrow (2)$$

which may be written as $L^T X = Y \rightarrow (3)$

$$(2) \ \& \ (3) \Rightarrow LY = b \rightarrow (4)$$

The immediate solution values y_i , $i = 1(1)n$ are obtained from (4) by forward substitution and the solution values x_i , $i = 1(1)n$ are determined from (3) by back substitution.

Alternatively, we can find only one inverse L^{-1} and obtain $Y = L^{-1}b$

$$\text{and } X = (L^T)^{-1}Y = (L^{-1})^T Y$$

This results in substantial saving of computational work. The inverse of A may be obtained from

$$A^{-1} = (L^{-1})^T L^{-1}$$

The elements of the lower triangular matrix L may be obtained as:

$$l_{ii} = \left(a_{ii} - \sum_{j=1}^{i-1} l_{ij}^2 \right)^{1/2}, \quad i = 1, 2, \dots, n$$

$$l_{ij} = \frac{1}{l_{ii}} \left(a_{ij} - \sum_{k=1}^{j-1} l_{jk} \cdot l_{ik} \right)$$

where $i = j+1, j+2, \dots, n$; $j = 1(1)n$

and $l_{ij} = 0$, $i < j$

Positive definite matrix:-

A symmetric matrix A is positive definite if and only if $x^T A x > 0$ for every column $x \neq 0$ in $\mathbb{R}^n$.

&

A square matrix is called positive definite if it is symmetric & all its eigenvalues are positive.

① Solve the system of equations using Cholesky method

$$x_1 + 2x_2 + 3x_3 = 5$$

$$2x_1 + 8x_2 + 22x_3 = 6$$

$$3x_1 + 22x_2 + 82x_3 = -10$$

Soln: The given system of equations can be written as:

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 8 & 22 \\ 3 & 22 & 82 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 6 \\ -10 \end{bmatrix}$$

Let $A = LL^T$

$$\text{i.e. } \begin{bmatrix} 1 & 2 & 3 \\ 2 & 8 & 22 \\ 3 & 22 & 82 \end{bmatrix} = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} l_{11} & l_{21} & l_{31} \\ 0 & l_{22} & l_{32} \\ 0 & 0 & l_{33} \end{bmatrix}$$

$$= \begin{bmatrix} l_{11}^2 & l_{11}l_{21} & l_{11}l_{31} \\ l_{21}l_{11} & l_{21}^2 + l_{22}^2 & l_{21}l_{31} + l_{22}l_{32} \\ l_{31}l_{11} & l_{31}l_{21} + l_{32}l_{22} & l_{31}^2 + l_{32}^2 + l_{33}^2 \end{bmatrix}$$

Comparing the corresponding elements on both sides,

we get: $l_{11}^2 = 1 \Rightarrow l_{11} = 1$

$$l_{11}l_{21} = 2 \Rightarrow l_{21} = 2$$

$$l_{11} l_{31} = 3 \Rightarrow l_{31} = 3$$

$$l_{21}^2 + l_{22}^2 = 8 \Rightarrow l_{22} = 2$$

$$l_{31} l_{21} + l_{32} l_{22} = 22 \Rightarrow l_{32} = 8$$

$$l_{31}^2 + l_{32}^2 + l_{33}^2 = 82 \Rightarrow l_{33} = 3$$

$$\text{Hence, } L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & 0 \\ 3 & 8 & 3 \end{bmatrix}$$

Now, we write $AX = b$ as $LL^T X = b$

Replacing $L^T X$ by y , we write $LY = b$

$$\text{i.e. } \begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & 0 \\ 3 & 8 & 3 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 6 \\ -10 \end{bmatrix} \Rightarrow \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} +5 \\ -2 \\ -3 \end{bmatrix}$$

Since $L^T X = y$

$$\text{i.e. } \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 8 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \\ -3 \end{bmatrix}$$

$$\text{Thus, } \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix}$$

② Solve the system of equations using Cholesky method.
Also, determine A^{-1} .

$$4x_1 - x_2 = 1$$

$$-x_1 + 4x_2 - x_3 = 0$$

$$-x_2 + 4x_3 - x_4 = 0$$

$$-x_3 + 4x_4 = 0$$

Soln: The given system of equations can be written as:-

$$\begin{bmatrix} 4 & -1 & 0 & 0 \\ -1 & 4 & -1 & 0 \\ 0 & -1 & 4 & -1 \\ 0 & 0 & -1 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Let $A = LL^T$

$$\text{i.e. } \begin{bmatrix} 4 & -1 & 0 & 0 \\ -1 & 4 & -1 & 0 \\ 0 & -1 & 4 & -1 \\ 0 & 0 & -1 & 4 \end{bmatrix} = \begin{bmatrix} l_{11} & 0 & 0 & 0 \\ l_{21} & l_{22} & 0 & 0 \\ l_{31} & l_{32} & l_{33} & 0 \\ l_{41} & l_{42} & l_{43} & 0 \end{bmatrix} \begin{bmatrix} l_{11} & l_{21} & l_{31} & l_{41} \\ 0 & l_{22} & l_{32} & l_{42} \\ 0 & 0 & l_{33} & l_{43} \\ 0 & 0 & 0 & l_{44} \end{bmatrix}$$

$$= \begin{bmatrix} l_{11}^2 & l_{11}l_{21} & l_{11}l_{31} & l_{11}l_{41} \\ l_{21}l_{11} & l_{21}^2 + l_{22}^2 & l_{21}l_{31} + l_{22}l_{32} & l_{21}l_{41} + l_{22}l_{42} \\ l_{31}l_{11} & l_{31}l_{21} + l_{32}l_{22} & l_{31}^2 + l_{32}^2 + l_{33}^2 & l_{31}l_{41} + l_{32}l_{42} + l_{33}l_{43} \\ l_{41}l_{11} & l_{41}l_{21} + l_{42}l_{22} & l_{41}l_{31} + l_{42}l_{32} + l_{43}l_{33} & l_{41}^2 + l_{42}^2 + l_{43}^2 + l_{44}^2 \end{bmatrix}$$

Comparing the corresponding elements on both sides,
we get: $l_{11} = 2$, $l_{21} = -1/2$, $l_{31} = 0$, $l_{41} = 0$

$$l_{21}^2 + l_{22}^2 = 4 \Rightarrow l_{22} = \sqrt{15/4}$$

$$l_{21}l_{31} + l_{22}l_{32} = -1 \Rightarrow l_{32} = -\sqrt{4/15}$$

$$l_{21}l_{41} + l_{22}l_{42} = 0 \Rightarrow l_{42} = 0$$

$$l_{31}^2 + l_{32}^2 + l_{33}^2 = 4 \Rightarrow l_{33} = \sqrt{56/15}$$

$$l_{31}l_{41} + l_{32}l_{42} + l_{33}l_{43} = -1 \Rightarrow l_{43} = -\sqrt{15/56}$$

$$l_{41}^2 + l_{42}^2 + l_{43}^2 + l_{44}^2 = 4 \Rightarrow l_{44} = \sqrt{\frac{209}{56}}$$

$$\text{Hence } L = \begin{bmatrix} 2 & 0 & 0 & 0 \\ -1/2 & \sqrt{15/4} & 0 & 0 \\ 0 & -\sqrt{4/15} & \sqrt{56/15} & 0 \\ 0 & 0 & -\sqrt{15/56} & \sqrt{\frac{209}{56}} \end{bmatrix}$$

Now, we write $Ax = b$ as $LL^T x = b$

Replacing $L^T x$ by y , we write $Ly = b$

$$\text{i.e. } \begin{bmatrix} 2 & 0 & 0 & 0 \\ -1/2 & \sqrt{15/4} & 0 & 0 \\ 0 & -\sqrt{4/15} & \sqrt{56/15} & 0 \\ 0 & 0 & -\sqrt{15/56} & \sqrt{\frac{209}{56}} \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} = \begin{bmatrix} 1/2 \\ 1/\sqrt{60} \\ 1/\sqrt{840} \\ 1/\sqrt{11704} \end{bmatrix}$$

Since $L^T X = Y$

$$\text{i.e.} \begin{bmatrix} 2 & -1/2 & 0 & 0 \\ 0 & \sqrt{15}/4 & -\sqrt{4}/15 & 0 \\ 0 & 0 & \sqrt{56}/15 & -\sqrt{15}/56 \\ 0 & 0 & 0 & \sqrt{209}/56 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1/2 \\ 1/\sqrt{60} \\ 1/\sqrt{840} \\ 1/\sqrt{11704} \end{bmatrix}$$

Thus, $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 56/209 \\ 15/209 \\ 4/209 \\ 1/209 \end{bmatrix}$

Now, to find A^{-1} :

w.k.t $A = LL^T$

$$\begin{aligned} \therefore A^{-1} &= (LL^T)^{-1} \\ &= (L^T)^{-1} L^{-1} \\ &= (L^{-1})^T L^{-1} \end{aligned}$$

Thus, we need to find L^{-1} .

$$L^{-1} = \begin{bmatrix} 1/2 & 0 & 0 & 0 \\ 1/\sqrt{60} & \sqrt{4/15} & 0 & 0 \\ 1/\sqrt{840} & \sqrt{2/105} & \sqrt{15/56} & 0 \\ 1/\sqrt{11704} & \sqrt{2/1463} & \sqrt{15/11704} & \sqrt{56/209} \end{bmatrix}$$

$$= \begin{bmatrix} 0.5 & 0 & 0 & 0 \\ 0.1291 & 0.5164 & 0 & 0 \\ 0.0345 & 0.1380 & 0.5176 & 0 \\ 0.0092 & 0.0370 & 0.1387 & 0.5176 \end{bmatrix}$$

Hence, $A^{-1} = (L^{-1})^T L^{-1} = \begin{bmatrix} 0.2679 & 0.0718 & 0.0191 & 0.0048 \\ 0.0718 & 0.2871 & 0.0766 & 0.0192 \\ 0.0191 & 0.0766 & 0.2871 & 0.0718 \\ 0.0048 & 0.0192 & 0.0718 & 0.2679 \end{bmatrix}$

Least Squares Solution for Inconsistent Systems.

Let the system of linear equations be $Ax = b \rightarrow (*)$ where A, x and b are of order $m \times n, n \times 1$ & $m \times 1$ respectively. Here, since the system is inconsistent, the system has no solution. Again, since there may be more than one x_1 (least-squares solutions) for which $\|Ax - b\|$ is minimum, there exist one such x_1 (say x_m) whose norm is minimum. That is, x_m is called minimum norm least squares solution if $\|x_m\| \leq \|x_1\|$ for any x_1 such that $\|Ax_1 - b\| \leq \|Ax - b\|$ for all x .

The minimum norm least squares solution can be determined using the relation $x = A^+ b$

Let A be a matrix of order $m \times n$. Then:

(i) If rank of A is 0, then A^+ is a null matrix of order $m \times m$

(ii) If rank of A is 1, then $A^+ = \frac{1}{\text{trace}(AA^t)} A^t$

(iii) If rank of A is n , then $A^+ = (A^t A)^{-1} A^t$

(iv) If rank of A is m , then $A^+ = A^t (AA^t)^{-1}$

where A^+ is the Moore-Penrose inverse (g-inverse) of A .

Since A^+ is unique, the minimum norm least squares solution is also unique.

The least squares solution can also be determined by the following way.

The vector $Ax - b$ in terms of elements of A , x & b is:-

$$\begin{bmatrix} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n - b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n - b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n - b_m \end{bmatrix}$$

Let square of $\|Ax - b\|$ be denoted by S . Then:

$$\begin{aligned} S &= (a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n - b_1)^2 + (a_{21}x_1 + a_{22}x_2 + \dots \\ &+ a_{2n}x_n - b_2)^2 + \dots + (a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n - b_m)^2 \\ &= \sum_{i=1}^m \sum_{j=1}^n (a_{ij}x_j - b_i)^2 \end{aligned}$$

Also, S is called the sum of square of residues.

The conditions for S to be minimum are

$$\frac{\partial S}{\partial x_1} = 0, \quad \frac{\partial S}{\partial x_2} = 0, \quad \dots, \quad \frac{\partial S}{\partial x_n} = 0.$$

This method is not suitable for a large system of equations, while the method stated earlier is applicable for a large system also.

① Find the least squares solution of the following equations: $x+y=3.0$, $2x-y=0.03$
 $x+3y=7.03$, $3x+y=4.97$

Also, estimate the residue.

Solⁿ: Let x, y be least squares solⁿ of given system.

Then the square of residues S is:

$$S = (x+y-3.0)^2 + (2x-y-0.03)^2 + (x+3y-7.03)^2 + (3x+y-4.97)^2$$

We choose x and y in such a way that S is minimum.

$$\therefore \frac{\partial S}{\partial x} = 0 \text{ and } \frac{\partial S}{\partial y} = 0.$$

That is:

$$2(x+y-3.0) + 4(2x-y-0.03) + 2(x+3y-7.03) + 6(3x+y-4.97) = 0$$

$$\& 2(x+y-3.0) - 2(2x-y-0.03) + 6(x+3y-7.03) + 2(3x+y-4.97) = 0$$

These equations reduce to:

$$3x+y-5=0 \text{ and } 5x+12y-29.03=0.$$

Solution of these eq^{ns} is $x = \frac{30.97}{31} = 0.9990322$ and

$$y = \frac{62.09}{31} = 2.0029032$$

which is required solⁿ of the given eq^{ns}.

The sum of the squares of residues is

$$S = (3.0019354 - 3.0)^2 + (-0.0048388 - 0.03)^2 + (7.0071418 - 7.03)^2 + (4.9999998 - 4.97)^2 = 0.0026129$$

② Find g-inverse of the singular matrix $A = \begin{bmatrix} 3 & 6 \\ 2 & 4 \end{bmatrix}$ and hence find a least squares solution of the inconsistent system : $3x + 6y = 9$.
 $2x + 4y = 5$.

Solⁿ:- To find g-inverse of A, first let us find Rank of A.

$$R_2: 3R_2 - 2R_1 \text{ leads us to } A = \begin{bmatrix} 3 & 6 \\ 0 & 0 \end{bmatrix}$$

$$\text{Thus, } \rho(A) = 1.$$

$$\therefore A^+ = \frac{1}{\text{trace}(AA^t)} \cdot A^t$$

$$AA^t = \begin{bmatrix} 3 & 6 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 6 & 4 \end{bmatrix} = \begin{bmatrix} 45 & 30 \\ 30 & 20 \end{bmatrix}$$

$$\therefore \text{trace of } AA^t = 65$$

$$\Rightarrow A^+ = \frac{1}{65} \begin{bmatrix} 3 & 2 \\ 6 & 4 \end{bmatrix}$$

Hence, least square solⁿ for $Ax = b$ is $x = A^+b$

$$\text{i.e. } x = \frac{1}{65} \begin{bmatrix} 3 & 2 \\ 6 & 4 \end{bmatrix} \begin{bmatrix} 9 \\ 5 \end{bmatrix} = \frac{1}{65} \begin{bmatrix} 37 \\ 74 \end{bmatrix}$$

$$\Rightarrow x = \frac{37}{65} \text{ and } y = \frac{74}{65}$$

Exercise:-

1) Find the Inverse of the matrix $A = \begin{bmatrix} 3 & 3 & 4 \\ 2 & 1 & 1 \\ 1 & 3 & 5 \end{bmatrix}$ using the matrix partition method.

Hence find the solution of the system of equations:

$$3x_1 + 3x_2 + 4x_3 = 5.$$

$$2x_1 + x_2 + x_3 = 7.$$

$$x_1 + 3x_2 + 5x_3 = 6.$$

Ans:- $A^{-1} = \begin{bmatrix} -2 & 3 & 1 \\ 9 & -11 & -5 \\ -5 & 6 & 3 \end{bmatrix}$

$$x_1 = 17, \quad x_2 = -62, \quad x_3 = 35.$$

2) Solve the following system of equations by Cholesky method:

$$2x_1 + x_2 - x_3 = 6$$

$$x_1 - 3x_2 + 5x_3 = 11$$

$$-x_1 + 5x_2 + 4x_3 = 13$$

Ans:- $x_1 = 1, \quad x_2 = 2, \quad x_3 = 2$

3) Obtain the g-inverse (Moore-Penrose inverse) of

$$\begin{bmatrix} 2 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 1 & 2 \end{bmatrix}$$

and use it to solve the following system of eq^{ns}:-

$$2x_1 + x_2 + x_4 = 4.$$

$$x_1 - x_3 = 0.$$

$$x_2 + x_3 + 2x_4 = 4.$$

Ans: $A^+ = \begin{bmatrix} 7/12 & -1/4 & -1/3 \\ 1/4 & -1/4 & 0 \\ 7/12 & -5/4 & -1/3 \\ -5/12 & 3/4 & 2/3 \end{bmatrix}$

$$x_1 = 1, x_2 = 1, x_3 = 1, x_4 = 1$$